Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which statements about the `final` modifier are correct? (Choose all that apply.)
   - **A.** Instance and `static` variables can be marked `final`.
   - **B.** A variable is effectively final only if it is marked `final`.
   - **C.** An object that is marked `final` cannot be modified.
   - **D.** Local variables cannot be declared with type `var` and the `final` modifier.
   - **E.** A primitive that is marked `final` cannot be modified.

2. Which of the following can fill in the blank in this code to make it compile? (Choose all that apply.)

```
public class Ant {
   _____ void method() {}
}
```

   - **A.** `default`
   - **B.** `final`
   - **C.** `private`
   - **D.** `Public`
   - **E.** `String`
   - **F.** `zzz:`

final compila y significa que el metodo no puede ser sobrescrito
private tambien compila y significa que solo se puede llamar al metodo desde la clase Ant

3. Which of the following methods compile? (Choose all that apply.)
   - **A.** `final static void rain() {}`
   - **B.** `public final int void snow() {}`
   - **C.** `private void int hail() {}`
   - **D.** `static final void sleet() {}`
   - **E.** `void final ice() {}`
   - **F.** `void public slush() {}`

A es correcta porque sigue la sintaxis tiene los especificadores(opcional) tipo de retorno, nombre del metodo parentesis y llaves

D es correcta porque tiene los especificadores el tipo de retorno, nombre del metodo, parentesis y llaves.

4. Which of the following can fill in the blank and allow the code to compile? (Choose all that apply.)

```
final ________ song = 6;
```

   - **A.** `int`
   - **B.** `Integer`
   - **C.** `long`
   - **D.** `Long`
   - **E.** `double`
   - **F.** `Double`

5. Which of the following methods compile? (Choose all that apply.)

A. `public void january() { return; }`

B. `public int february() { return null;}`

C. `public void march() {}`

D. `public int april() { return 9;}`

E. `public int may() { return 9.0;}`

F. `public int june() { return;}`

A es correcto porque retorna void y el return vacio es valido
C es correcto porque retorna void es correcto que no tenga record
D es correcto porque retorna un int que seria el 9

6. Which of the following methods compile? (Choose all that apply.)

A. `public void violin(int... nums) {}`

B. `public void viola(String values, int... nums) {}`

C. `public void cello(int... nums, String values) {}`

D. `public void bass(String... values, int... nums) {}`

E. `public void flute(String[] values, ...int nums) {}`

F. `public void oboe(String[] values, int[] nums) {}`

A esta bien su sintaxis usando Varargs ya que es unico, En B tambien esta bien ya que primero esta el parametro normal y al final el Varaargs y F es correcta porque son arreglos normales.

7. Given the following method, which of the method calls return 2? (Choose all that apply.)

```
public int juggle(boolean b, boolean... b2) {
   return b2.length;
}
```

A. `juggle();`

B. `juggle(true);`

C. `juggle(true, true);`

D. `juggle(true, true, true);`

E. `juggle(true, {true, true});`

F. `juggle(true, new boolean[2]);`

D es correcta porque b= true b2=true,true entonces b2.length es = 2
F es correcta porque b=true, b2 es de tamaño 2 entonces retorna 2

8. Which of the following statements is correct?

A. Package access is more lenient than `protected` access.

B. A `public` class that has private fields and package methods is not visible to classes outside the package.

C. You can use access modifiers so only some of the classes in a package see a particular package class.

D. You can use access modifiers to allow access to all methods and not any instance variables.

E. You can use access modifiers to restrict access to all classes that begin with the word `Test`.

D es la correcta porque se pueden usar poniendo los metodos public y las variables private

9. Given the following class definitions, which lines in the `main()` method generate a compiler error? (Choose all that apply.)

```
// Classroom.java
package my.school;
public class Classroom {
   private int roomNumber;
   protected static String teacherName;
   static int globalKey = 54321;
   public static int floor = 3;
   Classroom(int r, String t) {
      roomNumber = r;
      teacherName = t; } }
```

```
// School.java
1: package my.city;
2: import my.school.*;
3: public class School {
4:    public static void main(String[] args) {
5:       System.out.println(Classroom.globalKey);
6:       Classroom room = new Classroom(101, "Mrs. Anderson");
7:       System.out.println(room.roomNumber);
8:       System.out.println(Classroom.floor);
9:       System.out.println(Classroom.teacherName); } }
```

A. None: the code compiles fine.

B. Line 5

C. Line 6

D. Line 7

E. Line 8

F. Line 9

Linea 5 genera error porque globalKey es de acceso en el paquete static e int no son visibles en my.city
Linea 6 porque el constructor es de acceso en el paquete no es visible en my.city
Linea 7 roomNumber es private asi que no es visible desde cualquier clase
Linea 9 teacherName es protected asi que solo es visible dentro del paquete.

10. What is the output of executing the `Chimp` program?

```
// Rope.java
package rope;
public class Rope {
   public static int LENGTH = 5;
   static {
      LENGTH = 10;
   }
   public static void swing() {
```

```
      System.out.print("swing ");
   } }
```

```
// Chimp.java
1: import rope.*;
2: import static rope.Rope.*;
3: public class Chimp {
4:    public static void main(String[] args) {
5:       Rope.swing();
6:       new Rope().swing();
7:       System.out.println(LENGTH);
8:    } }
```

Rope.swing() imprime swing
new Rope().swing imprime swing y
System.out.println(LENGTH) imprime 10

A. `swing swing 5`

B. `swing swing 10`

C. Compiler error on line 2 of `Chimp`

D. Compiler error on line 5 of `Chimp`

E. Compiler error on line 6 of `Chimp`

F. Compiler error on line 7 of `Chimp`

11. Which statements are true of the following code? (Choose all that apply.)

```
1:  public class Rope {
2:     public static void swing() {
3:        System.out.print("swing");
4:     }
5:     public void climb() {
6:        System.out.println("climb");
7:     }
8:     public static void play() {
9:        swing();
10:       climb();
11:    }
12:    public static void main(String[] args) {
13:       Rope rope = new Rope();
14:       rope.play();
15:       Rope rope2 = null;
16:       System.out.print("-");
17:       rope2.play();
18:    } }
```

Error en la linea 10 ya que un metodo estatico no puede llamar a un metodo de instancia sin antes haber creado un objeto.

- **A.** The code compiles as is.
- **B.** There is exactly one compiler error in the code.
- **C.** There are exactly two compiler errors in the code.
- **D.** If the line(s) with compiler errors are removed, the output is `swing-climb`.
- **E.** If the line(s) with compiler errors are removed, the output is `swing-swing`.
- **F.** If the line(s) with compile errors are removed, the code throws a `NullPointerException`.

12. How many variables in the following method are effectively final?

```
public void feed() {
   int monkey = 0;
   if(monkey > 0) {
      var giraffe = monkey++;
      String name;
      name = "geoffrey";
   }
   String name = "milly";
   var food = 10;
   while(monkey <= 10) {
      food = 0;
   }
   name = null;
}
```

Ambas se les asigna un valor y nunca cambia

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4
- **E.** 5
- **F.** None of the above. The code does not compile.

13. What is the output of the following code?

```
// RopeSwing.java
import rope.*;
import static rope.Rope.*;
public class RopeSwing {
   private static Rope rope1 = new Rope();
   private static Rope rope2 = new Rope();
   {
      System.out.println(rope1.length);
   }
```

```
   public static void main(String[] args) {
      rope1.length = 2;
      rope2.length = 8;
      System.out.println(rope1.length);
   }
}

// Rope.java
package rope;
public class Rope {
   public static int length = 0;
}
```

A. 02

B. 08

C. 2

D. 8

E. The code does not compile.

F. An exception is thrown.

14. How many lines in the following code have compiler errors?

```
1:  public class RopeSwing {
2:     private static final String leftRope;
3:     private static final String rightRope;
4:     private static final String bench;
5:     private static final String name = "name";
6:     static {
7:        leftRope = "left";
8:        rightRope = "right";
9:     }
10:    static {
11:       name = "name";
12:       rightRope = "right";
13:    }
14:    public static void main(String[] args) {
15:       bench = "bench";
16:    }
17: }
```

Linea 4 a bench no se le asigna en el bloque estatico
Linea 11 name ya fue asignada en la linea 5
Linea 12 porque ya fue asignada en la linea 8
Linea 15 porque no se le puede asignar static final dentro de un metodo.

A. 0

B. 1

C. 2
D. 3
E. 4
F. 5

15. Which of the following can replace line 2 to make this code compile? (Choose all that apply.)

```
1: import java.util.*;
2: // INSERT CODE HERE
3: public class Imports {
4:    public void method(ArrayList<String> list) {
5:       sort(list);
6:    }
7: }
```

A. `import static java.util.Collections;`
B. `import static java.util.Collections.*;`
C. `import static java.util.Collections.sort(ArrayList<String>);`
D. `static import java.util.Collections;`
E. `static import java.util.Collections.*;`
F. `static import java.util.Collections.sort(ArrayList<String>);`

16. What is the result of the following statements?

```
public class Test {
   public void print(byte x) {
      System.out.print("byte-");
   }
   public void print(int x) {
      System.out.print("int-");
   }
   public void print(float x) {
      System.out.print("float-");
   }
   public void print(Object x) {
      System.out.print("Object-");
   }
   public static void main(String[] args) {
      Test t = new Test();
      short s = 123;
      t.print(s);
      t.print(true);
```

```
      t.print(6.789);
   }
}
```

- **A.** `byte-float-Object-`
- **B.** `int-float-Object-`
- **C.** `byte-Object-float-`
- **D.** `int-Object-float-`
- **E.** `int-Object-Object-`
- **F.** `byte-Object-Object-`

17. What is the result of the following program?

```
1:  public class Squares {
2:     public static long square(int x) {
3:        var y = x * (long) x;
4:        x = -1;
5:        return y;
6:     }
7:     public static void main(String[] args) {
8:        var value = 9;
9:        var result = square(value);
10:       System.out.println(value);
11:    } }
```

Java pasa primitivos por valor al llamar a square(value) se copia el valor de value y ya al final se imprime value que seria 9.

- **A.** `-1`
- **B.** `9`
- **C.** `81`
- **D.** Compiler error on line 9
- **E.** Compiler error on a different line

18. Which of the following are output by the following code? (Choose all that apply.)

```
public class StringBuilders {
   public static StringBuilder work(StringBuilder a,
      StringBuilder b) {
      a = new StringBuilder("a");
      b.append("b");
      return a;
   }
   public static void main(String[] args) {
      var s1 = new StringBuilder("s1");
      var s2 = new StringBuilder("s2");
```

```
        var s3 = work(s1, s2);
        System.out.println("s1 = " + s1);
        System.out.println("s2 = " + s2);
        System.out.println("s3 = " + s3);
    }
}
```

Dentro del metodo a = new StringBuilder("a") se crea un objeto nuevo pero sigue apuntando a s1
b.append("b") modifica el objeto origial ya que b y s2 apuntan al mismo objeto
El metodo retorna a.

- **A.** `s1 = a`
- **B.** `s1 = s1`
- **C.** `s2 = s2`
- **D.** `s2 = s2b`
- **E.** `s3 = a`
- **F.** The code does not compile.

19. Which of the following will compile when independently inserted in the following code? (Choose all that apply.)

```
1:  public class Order3 {
2:     final String value1 = "red";
3:     static String value2 = "blue";
4:     String value3 = "yellow";
5:     {
6:        // CODE SNIPPET 1
7:     }
8:     static {
9:        // CODE SNIPPET 2
10:    } }
```

value1 es estatica entonces es accesible en el bloque
value3 es de instancia entonces es accesible desde el bloque
value2 es estatica y accesible desde el bloque.

- **A.** Insert at line 6: `value1 = "green";`
- **B.** Insert at line 6: `value2 = "purple";`
- **C.** Insert at line 6: `value3 = "orange";`
- **D.** Insert at line 9: `value1 = "magenta";`
- **E.** Insert at line 9: `value2 = "cyan";`
- **F.** Insert at line 9: `value3 = "turquoise";`

20. Which of the following are true about the following code? (Choose all that apply.)

```
public class Run {
    static void execute() {
        System.out.print("1-");
    }
    static void execute(int num) {
        System.out.print("2-");
    }
```

```
    static void execute(Integer num) {
        System.out.print("3-");
    }
    static void execute(Object num) {
        System.out.print("4-");
    }
    static void execute(int... nums) {
        System.out.print("5-");
    }
    public static void main(String[] args) {
        Run.execute(100);
        Run.execute(100L);
    }
}
```

- **A.** The code prints out `2-4-`.
- **B.** The code prints out `3-4-`.
- **C.** The code prints out `4-2-`.
- **D.** The code prints out `4-4-`.
- **E.** The code prints `3-4-` if you remove the method `static void execute(int num)`.
- **F.** The code prints `4-4-` if you remove the method `static void execute(int num)`.

21. Which method signatures are valid overloads of the following method signature? (Choose all that apply.)

`public void moo(int m, int... n)`

- **A.** `public void moo(int a, int... b)`
- **B.** `public int moo(char ch)`
- **C.** `public void moooo(int... z)`
- **D.** `private void moo(int... x)`
- **E.** `public void moooo(int y)`
- **F.** `public void moo(int... c, int d)`
- **G.** `public void moo(int... i, int j...)`

B diferente nombre y diferente lista de parametros
D mismo nombre pero diferente lista de parametros.